EVOCURE Candidate Screening

Pancreatic ductal adenocarcinoma, five prioritised compounds

EVOCURE-PDAC-20261027 | generated 2026-10-29 13:39:36 UTC

What this document is

A demonstration of an open source in silico screening platform. It shows what the software computed on one reproducible run: a bounded evolutionary search over molecular structures, scored by a QSAR model, toxicology heuristics and a simulated two dimensional tumour microenvironment.

What it is not: evidence that any compound here treats pancreatic cancer. No molecule in this report has been synthesised, assayed in a cell, or given to an animal or a person. These are computational hypotheses at the earliest possible stage of discovery, and the large majority of such hypotheses fail.

Result in one line

5 of 5 candidates separated from the negative control at their point estimates, and every one of their potency intervals also admits no effect at all.

The screening pipeline ran end to end and produced five ranked candidates. The tissue assay was checked with a positive and a negative control before the candidates were read, so the result above is interpretable rather than merely generated. The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

Run at a glance

Structures evaluated	467 unique valid structures
Generations	20, population 24
Replicate seeds per compound	5
Compute backend	NumPy CPU float64
QSAR training set	43 curated compounds
QSAR scaffold split R2	0.699
QSAR classifier ROC AUC	0.954
Wall clock	1386 seconds

Open source platform and data. Reproduce this document with the commands in the reproducibility section. Report generated directly from the run's results file; no figure in it was entered by hand.

1. Summary

A bounded evolutionary search started from four approved pancreatic cancer drugs and generated structural variants through point mutations and scaffold recombination. Every variant was validated with RDKit, scored on five objectives, and the non dominated front was carried into the next generation. The five compounds below are the highest ranked variants that were not themselves seeds.

1.1 Assay validity, checked before reading the candidates

Role	Compound	TGI %	95% CI	CI excludes 0
Positive, curated	Gemcitabine	67.1	60.4 to 73.8	yes
Negative	Caffeine	-5.7	-18.3 to 6.8	no
Positive, modelled	Gemcitabine	-1.0	-14.7 to 12.7	no

The third arm is the same positive control molecule with its curated bioactivity withheld, so its potency comes from the structure-property heuristic. Note that the candidates in this run are scored by the calibrated QSAR instead, so this arm is no longer a like for like comparator for them. It still shows how far a potency source can move the same molecule: 179 fold here, from separating cleanly to not separating at all.

Does the simulated assay work?
Positive versus negative control: $p < 0.001$, mean difference 72.9 percentage points, Hedges g 8.12.

The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

1.1a The comparison is between a measurement and a prediction

The positive control fails its own test when scored like a candidate
Gemcitabine draws its potency from curated experimental bioactivity: IC50 0.0450 uM, and it inhibits growth by 67.1 %. A generated structure has no measurement, so it is scored by the structure-property model. Scoring this same molecule that way gives IC50 8.053 uM, 179 times weaker, and growth inhibition falls to -1.0 %, which does not separate from the negative control.

Nothing about the molecule changed. Only the provenance of its potency did. Any candidate in this report is therefore being compared against a number no generated structure can ever be assigned, and failing that comparison says little about the chemistry.

1.2 The five candidates

ID	Compound	pIC50	Score	TGI %	95% CI	vs control
Candidate 01	Compound-A45A80	6.55	0.656	63.8	52.3 to 75.2	separates, corrected
Candidate 02	Compound-86B823	6.50	0.654	64.3	49.7 to 79.0	separates, corrected
Candidate 03	Compound-4376BB	6.97	0.652	71.1	61.2 to 81.1	separates, corrected
Candidate 04	Compound-677FA4	7.01	0.652	76.3	63.6 to 88.9	separates, corrected
Candidate 05	Compound-A215BD	6.71	0.651	67.8	62.2 to 73.3	separates, corrected

TGI is tumour growth inhibition in the simulated microenvironment, relative to an untreated arm, averaged over independent seeds. Score is the weighted composite across the five search objectives. The last column reports Welch's t-test of each candidate against the negative control; it is the column that decides whether a point estimate means anything.

Every one of these separations also admits no effect

The table above reports each candidate at its point estimate of potency. Carrying the potency model's own 95% conformal interval through the same tissue simulation gives, for all 5 candidates, a band of responses that includes zero.

So the p values are real and they are not measuring what they appear to measure. They describe how reproducibly the simulation returns the same answer across seeds, given one assumed potency. They say nothing about whether that potency is right, and the model itself cannot rule out that these compounds do nothing.

Each candidate page shows its own band. This is the number to read before the p value, not after it.

What separation means when potency is predicted, not measured

Every candidate above draws its potency from the calibrated QSAR, not from a measurement. The tissue model then converts that potency into growth inhibition, so these results restate the QSAR's prediction through a simulation rather than testing it.

Two numbers decide how much weight that carries. The five candidates span a factor of 1.7 in assigned IC50. The QSAR's own 95% conformal interval is a factor of 1,330 on any single compound. The candidates are therefore ranked on differences far smaller than the uncertainty in any one of them, and the p values below describe the reproducibility of the simulation across seeds, not confidence in the underlying potency.

Section 2.5 shows the tissue model's full response to assigned potency, which is what these separations are made of.

Scope of the positive finding

Separated at an uncorrected $p < 0.05$: Candidate 01, Candidate 02, Candidate 03, Candidate 04, Candidate 05. 5 of those survived Bonferroni correction for 5 comparisons, which is the threshold that matters when several candidates are tested against one control arm.

Either way this is a result inside a simulation whose parameters are not experimentally calibrated. It justifies further computational scrutiny and nothing beyond that. Separation here is not potency, not selectivity, not safety and not evidence about human disease.

2. Method

2.1 Search

Objective prompt	Find 5 experimental candidates for pancreatic cancer based on Gemcitabine, 5-Fluorouracil, Erlotinib and Paclitaxel. Prioritize activity, low predicted toxicity and drug-likeness.
Planner	local_rule_based_planner
Derived priorities	safety, druglikeness, activity
Seed compounds	Gemcitabine, 5-Fluorouracil, Paclitaxel, Erlotinib
Generations	20
Population size	24
Applied concentration	5.0 uM
Exposure duration	24.0 hours
Search seed	20261027

The planner converts a plain language objective into search priorities only. It does not propose molecules and it does not supply any activity, potency or safety number. All quantitative output comes from the deterministic scientific models.

2.2 Objective weights

Objective	Weight	Source of the value
Predicted PDAC activity	2.0	Calibrated QSAR regressor, sigmoidal utility at pIC50 6.0
Low toxicological risk	2.0	Ten category structural alert and rule based risk index
Drug likeness (QED)	2.0	RDKit quantitative estimate of drug likeness
Selectivity index	1.0	ADMET derived tumour versus healthy cell window
Applicability domain	1.0	Maximum Tanimoto similarity to the training set
Simulated growth inhibition	1.0	Agent based tumour microenvironment, treated versus untreated

Tumour cell kill is never maximised alone. A compound that scores well on activity but triggers toxicity alerts or falls outside the applicability domain is penalised by construction.

2.2a What selection actually optimises

The selection loop is closed

Every candidate that could be dosed was simulated in the tumour microenvironment, 939 runs in total, and its growth inhibition entered the objective set used for Pareto sorting and survivor selection. The tumour model therefore exerts selective pressure on the population rather than only reporting on the winners.

Two limits on how much that is worth. All arms share one seed per generation, so the comparison is paired but the objective is the noisiest of the six. And the tissue model's drug response is driven by the predicted IC50, so this objective is correlated with the activity objective rather than independent of it. It adds diffusion, uptake, hypoxia and clonal structure on top of that prediction; it does not replace it with a measurement.

2.3 The QSAR model, and what it can carry

Training compounds	43
Independent scaffolds	32
Validation	repeated scaffold-grouped CV, 5 folds x 10 repeats
Pooled test predictions	430
Regression RMSE	1.340 log units

Regression R2	0.699
Classifier ROC AUC	0.954
Classifier Brier score	0.100
Conformal 95% half width	3.12 log units

Ordering is usable, individual values are not

Under repeated scaffold-grouped cross validation the regressor explains a real share of the variance across held out scaffolds (R2 = 0.70, RMSE 1.34 log units), and the classifier separates active from inactive well (ROC AUC = 0.95).

That does not make a predicted IC50 a quantity. The calibrated 95% conformal interval is about 3.1 log units in each direction, a factor of roughly 1,330 on IC50. Rank ordering within the generated set is defensible; a single predicted IC50 is not, and every pIC50 here is printed with its interval for that reason.

A note on method. These figures come from repeated grouped folds, not one split. A single scaffold split on 43 compounds leaves about ten test compounds and returns anything from R2 -0.09 to 0.82 depending only on the random seed, so a value from one split describes the split rather than the model.

2.4 Tissue simulation

Model	Two dimensional agent based PDAC microenvironment
Grid	36 x 28 cells over 800 x 600 micrometres
Fields	Oxygen and nutrient reaction diffusion, drug diffusion and uptake
Features	Hypovascularity, dense collagen matrix, sensitive and resistant clones
Comparison	Treated arm against an untreated arm with the same seed
Replicates	5 independent seeds per compound

Kinetics and microenvironment parameters are exploratory and have not been fitted to experimental pancreatic tumour data. The model's drug response is driven by the predicted IC50, so two compounds with the same predicted IC50 produce the same trajectory. Simulated growth inhibition is a property of this model, not a measurement.

2.5 What the tissue model contributes

Holding every pharmacology parameter fixed except the assigned IC50 isolates the tissue model's own contribution to telling compounds apart. The curve below is that contribution in full, at the dose used in this report.

Assigned IC50 (uM)	Growth inhibition %	SD across seeds
0.010	92.9	3.2
0.030	90.5	3.3
0.100	73.7	3.4
0.300	67.7	3.1
1.000	44.0	14.0
2.000	25.1	19.5
5.000	4.4	15.3
10.000	4.0	13.2
30.000	-6.3	10.9
100.000	-4.5	7.8

The ranking is finer than the measurement it rests on

At a fixed dose the response saturates well below the applied concentration and collapses above it, so growth inhibition behaves close to a threshold on whatever potency the model was handed.

The 5 candidates in this report span IC50 0.0660 to 0.1143 uM, a factor of 1.7. The potency model that produced those values carries a 95% conformal interval of a factor of 1,330. Ordering compounds by a 1.7 fold difference inside a 1330 fold uncertainty is not a ranking anyone should act on.

Reproduce with scripts/measure_tissue_transfer.py.

2.6 What the search explored, and what it discarded

This report shows 5 structures. The search evaluated 467, across 115 distinct Bemis-Murcko scaffolds. Everything else was scored and rejected, and the reasons are recorded rather than discarded.

Why a structure did not reach this report	Count	Share
ranked below the reporting cut, no single disqualifying criterion	256	54.8 %
medicinal chemistry alerts (BRENK)	103	22.1 %
medicinal chemistry alerts (BRENK, NIH)	58	12.4 %
low drug likeness	23	4.9 %
medicinal chemistry alerts (NIH)	13	2.8 %
low predicted activity	7	1.5 %
outside the model's applicability domain	1	0.2 %
medicinal chemistry alerts (BRENK, PAINS_B)	1	0.2 %

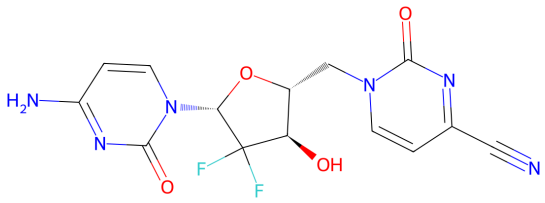
The generator explores further than the selection will follow

The search reached 115 distinct scaffolds, so it is not merely decorating the seed drugs. Selection then pulls back toward the training set, because the applicability domain objective rewards similarity to the 43 compounds the potency model was fitted on, and the model cannot speak confidently about anything further away.

The practical consequence is that the novelty ceiling is set by the training set, not by the generator. Improving the generator would not move it.

All 467 structures are kept in a local archive as a search trace, queryable with scripts/explore_archive.py. Re-scoring that archive against a better model takes about 99 seconds, against 6834 to repeat the search. It is a log of what an algorithm tried, not a library of candidates: scores come from a potency model whose interval is far wider than the differences between entries.

Candidate 01: Compound-A45A80



Identity

Canonical SMILES	N#Cc1ccn(C[C@H]2O[C@@H](n3ccc(N)nc3=O)C(F)(F)[C@@H]2O)c(=O)n1
InChIKey	ZSKCGNNRQRJSIF-FBIMIBRVSA-N
Molecular formula	C14H12F2N6O4
Murcko scaffold	O=c1ncccn1CC1CCC(n2cccnc2=O)O1
Origin	generation 20 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
366.3	-1.15	149.1	2	8	0.691	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	6.546	3.42 to 9.67	ordering signal
Implied IC50 (uM)	0.2845	spans orders of magnitude	not a quantity
Active probability	0.841	0.74 to 0.94	the stronger prediction
Applicability domain	0.67	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.633	0.000	0.691	0.916	0.667	0.656	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	63.75 % over 5 seeds
95% confidence interval	52.29 % to 75.22 %
Interval excludes zero	yes
Per seed values	77.1, 64.9, 61.9, 63.6, 51.2
Resistant clone fraction	0.660
Value used as selection objective	77.1 % on the search seed
Versus negative control	p < 0.001, Hedges g 6.48
Distinguishable, uncorrected	yes
Distinguishable, Bonferroni corrected	yes

Does something like this already exist?

Local verdict	Related to known chemistry, clearly distinct structure
Nearest local structure	Gemcitabine (Tanimoto 0.654)
Exact match	none in the local corpus
Same skeleton	none
Shared Murcko scaffold	none
Corpus searched	111 local structures

Closest local structure	Source	Tanimoto	Same scaffold
Gemcitabine	EVOCURE compound library	0.654	no
Capecitabine	Curated reference dataset	0.232	no
Capecitabine	EVOCURE compound library	0.195	no
Rosiglitazone	Curated reference dataset	0.165	no
Pioglitazone	EVOCURE compound library	0.157	no

This is not a prior art search

This searches roughly 120 local structures. It is a sanity check, not a prior art search. A structure absent here may still be published, catalogued or patented, and nothing in this result establishes that it is new. Confirming novelty requires PubChem, ChEMBL, SureChEMBL and a patent search, none of which run offline. Similarity to an active drug does not transfer that drug's activity, and distance from one is not evidence of anything.

The same simulation across the potency interval

Assigned IC50 (uM)	Growth inhibition %
0.0002	90.5
0.0078	88.7
0.2845	63.1
10.3724	-5.4
378.2077	-4.5

What the uncertainty actually costs

The potency above is a prediction, not a measurement, and its 95% conformal interval spans a factor of 1,767,665 in IC50. Running the same tissue model across that interval gives growth inhibition anywhere from -5.4 % to 90.5 %, against the 63.1 % reported at the point estimate.

The potency interval admits tissue responses from -5.4 to 90.5 percent growth inhibition. That band includes no effect, so the simulation does not establish that this compound does anything.

Conformal prediction gives a set with 95% coverage, not a distribution, so no mean or spread over this band is quoted: the method does not say where inside the interval the truth lies.

Could a chemist make this?

SA score	Band	Alerts	Families	Verdict
4.27	demanding	0	none	no alerts, synthetically reachable

Substructure alerts and SA score are coarse heuristics with known false positives; approved drugs trip them. Passing means not obviously implausible, not that the compound can be made or works.

Potential, and what it rests on

Signal	Reading	What it is based on
Predicted potency	favourable	pIC50 6.55, calibrated qsar
Potency certainty	low by construction	conformal interval 3.1 log units
Simulated tissue effect	distinguishable	TGI 63.8 % over 5 seeds, p < 0.001
Toxicology alerts	favourable	risk index 8.4 of 100, ten category screen
Drug likeness	favourable	QED 0.691, Lipinski passes
Model applicability	inside domain	Tanimoto 0.67 to nearest training compound
Structural distinctness	Related to known chemistry	max Tanimoto 0.65 over 111 local structures
Medicinal chemistry	no alerts	SA 4.27 (demanding), PAINS, Brenk and NIH catalogues

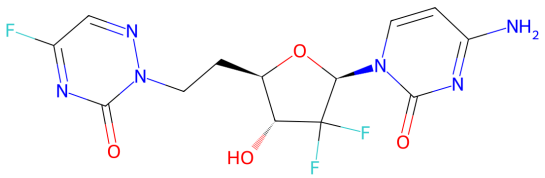
Not assessed at all

whether a synthetic route actually exists (SA score estimates difficulty, it does not plan a synthesis); chemical stability and reactivity under real conditions; absorption, distribution, metabolism and excretion in a living system; whether the compound reaches a pancreatic tumour through desmoplastic stroma; any experimental measurement whatsoever.

Verdict

This candidate separated from the negative control and survived correction for the number of candidates tested. That warrants further computational scrutiny and nothing more.

Candidate 02: Compound-86B823



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](CCn3ncc(F)nc3=O)[C@@H](O)C2(F)F)c(=O)n1
InChIKey	BBNOOVVXAYGCAT-BDODKLCJSA-N
Molecular formula	C13H13F3N6O4
Murcko scaffold	O=c1ncccn1C1CCC(CCn2nccnc2=O)O1
Origin	generation 20 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
374.3	-1.10	138.2	2	8	0.688	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	6.502	3.38 to 9.63	ordering signal
Implied IC50 (uM)	0.3145	spans orders of magnitude	not a quantity
Active probability	0.867	0.77 to 0.97	the stronger prediction
Applicability domain	0.69	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.623	0.000	0.688	0.916	0.685	0.654	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	64.34 % over 5 seeds
95% confidence interval	49.71 % to 78.97 %
Interval excludes zero	yes
Per seed values	77.1, 54.0, 64.3, 75.0, 51.2
Resistant clone fraction	0.653
Value used as selection objective	77.1 % on the search seed
Versus negative control	p < 0.001, Hedges g 5.76
Distinguishable, uncorrected	yes
Distinguishable, Bonferroni corrected	yes

Does something like this already exist?

Local verdict	Related to known chemistry, clearly distinct structure
Nearest local structure	Gemcitabine (Tanimoto 0.643)
Exact match	none in the local corpus
Same skeleton	none
Shared Murcko scaffold	none
Corpus searched	111 local structures

Closest local structure	Source	Tanimoto	Same scaffold
Gemcitabine	EVOCURE compound library	0.643	no
Capecitabine	Curated reference dataset	0.229	no
Capecitabine	EVOCURE compound library	0.207	no
Linezolid	EVOCURE compound library	0.159	no
Linezolid	Curated reference dataset	0.159	no

This is not a prior art search

This searches roughly 120 local structures. It is a sanity check, not a prior art search. A structure absent here may still be published, catalogued or patented, and nothing in this result establishes that it is new. Confirming novelty requires PubChem, ChEMBL, SureChEMBL and a patent search, none of which run offline. Similarity to an active drug does not transfer that drug's activity, and distance from one is not evidence of anything.

The same simulation across the potency interval

Assigned IC50 (uM)	Growth inhibition %
0.0002	90.5
0.0086	90.0
0.3145	60.9
11.4662	4.0
418.0898	-5.4

What the uncertainty actually costs

The potency above is a prediction, not a measurement, and its 95% conformal interval spans a factor of 1,767,665 in IC50. Running the same tissue model across that interval gives growth inhibition anywhere from -5.4 % to 90.5 %, against the 60.9 % reported at the point estimate.

The potency interval admits tissue responses from -5.4 to 90.5 percent growth inhibition. That band includes no effect, so the simulation does not establish that this compound does anything.

Conformal prediction gives a set with 95% coverage, not a distribution, so no mean or spread over this band is quoted: the method does not say where inside the interval the truth lies.

Could a chemist make this?

SA score	Band	Alerts	Families	Verdict
4.32	demanding	0	none	no alerts, synthetically reachable

Substructure alerts and SA score are coarse heuristics with known false positives; approved drugs trip them. Passing means not obviously implausible, not that the compound can be made or works.

Potential, and what it rests on

Signal	Reading	What it is based on
Predicted potency	favourable	pIC50 6.50, calibrated qsar
Potency certainty	low by construction	conformal interval 3.1 log units
Simulated tissue effect	distinguishable	TGI 64.3 % over 5 seeds, p < 0.001
Toxicology alerts	favourable	risk index 8.4 of 100, ten category screen
Drug likeness	favourable	QED 0.688, Lipinski passes
Model applicability	inside domain	Tanimoto 0.69 to nearest training compound
Structural distinctness	Related to known chemistry	max Tanimoto 0.64 over 111 local structures
Medicinal chemistry	no alerts	SA 4.32 (demanding), PAINS, Brenk and NIH catalogues

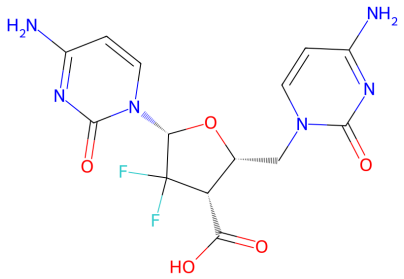
Not assessed at all

whether a synthetic route actually exists (SA score estimates difficulty, it does not plan a synthesis); chemical stability and reactivity under real conditions; absorption, distribution, metabolism and excretion in a living system; whether the compound reaches a pancreatic tumour through desmoplastic stroma; any experimental measurement whatsoever.

Verdict

This candidate separated from the negative control and survived correction for the number of candidates tested. That warrants further computational scrutiny and nothing more.

Candidate 03: Compound-4376BB



Identity

Canonical SMILES	Nc1ccn(C[C@H]2O[C@@H](n3ccc(N)nc3=O)C(F)(F)[C@H]2C(=O)O)c(=O)n1
InChIKey	MYWCKSMYURFBNR-KNBVPPQFSA-N
Molecular formula	C14H14F2N6O5
Murcko scaffold	O=c1ncccn1CC1CCC(n2cccnc2=O)O1
Origin	generation 20 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
384.3	-1.10	168.3	3	8	0.592	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	6.967	3.84 to 10.09	ordering signal
Implied IC50 (uM)	0.1079	spans orders of magnitude	not a quantity
Active probability	0.819	0.72 to 0.92	the stronger prediction
Applicability domain	0.67	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.725	0.000	0.592	0.916	0.667	0.652	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	71.12 % over 5 seeds
95% confidence interval	61.16 % to 81.07 %
Interval excludes zero	yes
Per seed values	74.3, 59.5, 66.7, 79.5, 75.6
Resistant clone fraction	0.684
Value used as selection objective	74.3 % on the search seed
Versus negative control	p < 0.001, Hedges g 7.60
Distinguishable, uncorrected	yes
Distinguishable, Bonferroni corrected	yes

Does something like this already exist?

Local verdict	Related to known chemistry, clearly distinct structure
Nearest local structure	Gemcitabine (Tanimoto 0.667)
Exact match	none in the local corpus
Same skeleton	none
Shared Murcko scaffold	none
Corpus searched	111 local structures

Closest local structure	Source	Tanimoto	Same scaffold
Gemcitabine	EVOCURE compound library	0.667	no
Capecitabine	Curated reference dataset	0.250	no
Capecitabine	EVOCURE compound library	0.210	no
Ampicillin	Curated reference dataset	0.184	no
Amoxicillin	EVOCURE compound library	0.179	no

This is not a prior art search

This searches roughly 120 local structures. It is a sanity check, not a prior art search. A structure absent here may still be published, catalogued or patented, and nothing in this result establishes that it is new. Confirming novelty requires PubChem, ChEMBL, SureChEMBL and a patent search, none of which run offline. Similarity to an active drug does not transfer that drug's activity, and distance from one is not evidence of anything.

The same simulation across the potency interval

Assigned IC50 (uM)	Growth inhibition %
0.0001	90.5
0.0030	87.7
0.1079	69.1
3.9345	7.0
143.4635	-4.5

What the uncertainty actually costs

The potency above is a prediction, not a measurement, and its 95% conformal interval spans a factor of 1,767,665 in IC50. Running the same tissue model across that interval gives growth inhibition anywhere from -4.5 % to 90.5 %, against the 69.1 % reported at the point estimate.

The potency interval admits tissue responses from -4.5 to 90.5 percent growth inhibition. That band includes no effect, so the simulation does not establish that this compound does anything.

Conformal prediction gives a set with 95% coverage, not a distribution, so no mean or spread over this band is quoted: the method does not say where inside the interval the truth lies.

Could a chemist make this?

SA score	Band	Alerts	Families	Verdict
4.19	demanding	0	none	no alerts, synthetically reachable

Substructure alerts and SA score are coarse heuristics with known false positives; approved drugs trip them. Passing means not obviously implausible, not that the compound can be made or works.

Potential, and what it rests on

Signal	Reading	What it is based on
Predicted potency	favourable	pIC50 6.97, calibrated qsar
Potency certainty	low by construction	conformal interval 3.1 log units
Simulated tissue effect	distinguishable	TGI 71.1 % over 5 seeds, p < 0.001
Toxicology alerts	favourable	risk index 8.4 of 100, ten category screen
Drug likeness	moderate	QED 0.592, Lipinski passes
Model applicability	inside domain	Tanimoto 0.67 to nearest training compound
Structural distinctness	Related to known chemistry	max Tanimoto 0.67 over 111 local structures
Medicinal chemistry	no alerts	SA 4.19 (demanding), PAINS, Brenk and NIH catalogues

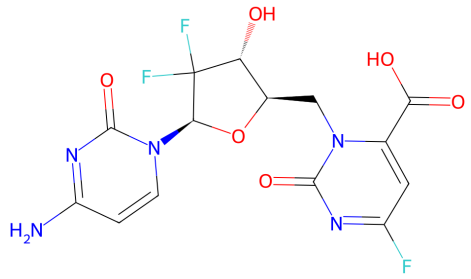
Not assessed at all

whether a synthetic route actually exists (SA score estimates difficulty, it does not plan a synthesis); chemical stability and reactivity under real conditions; absorption, distribution, metabolism and excretion in a living system; whether the compound reaches a pancreatic tumour through desmoplastic stroma; any experimental measurement whatsoever.

Verdict

This candidate separated from the negative control and survived correction for the number of candidates tested. That warrants further computational scrutiny and nothing more.

Candidate 04: Compound-677FA4



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](Cn3c(C(=O)O)cc(F)nc3=O)[C@@H](O)C2(F)F)c(=O)n1
InChIKey	LSNCQPWQYYMUOL-KNBVPPQFSA-N
Molecular formula	C14H12F3N5O6
Murcko scaffold	O=c1ncccn1CC1CCC(n2cccnc2=O)O1
Origin	generation 20 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
403.3	-1.19	162.6	3	8	0.533	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	7.013	3.89 to 10.14	ordering signal
Implied IC50 (uM)	0.0971	spans orders of magnitude	not a quantity
Active probability	0.892	0.79 to 0.99	the stronger prediction
Applicability domain	0.66	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.734	0.000	0.533	0.916	0.661	0.652	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	76.25 % over 5 seeds
95% confidence interval	63.62 % to 88.88 %
Interval excludes zero	yes
Per seed values	85.7, 62.2, 71.4, 86.4, 75.6
Resistant clone fraction	0.692
Value used as selection objective	85.7 % on the search seed
Versus negative control	p < 0.001, Hedges g 7.30
Distinguishable, uncorrected	yes
Distinguishable, Bonferroni corrected	yes

Does something like this already exist?

Local verdict	Related to known chemistry, clearly distinct structure
Nearest local structure	Gemcitabine (Tanimoto 0.621)
Exact match	none in the local corpus
Same skeleton	none
Shared Murcko scaffold	none
Corpus searched	111 local structures

Closest local structure	Source	Tanimoto	Same scaffold
Gemcitabine	EVOCURE compound library	0.621	no
Capecitabine	Curated reference dataset	0.238	no
Capecitabine	EVOCURE compound library	0.216	no
Linezolid	EVOCURE compound library	0.182	no
Linezolid	Curated reference dataset	0.182	no

This is not a prior art search

This searches roughly 120 local structures. It is a sanity check, not a prior art search. A structure absent here may still be published, catalogued or patented, and nothing in this result establishes that it is new. Confirming novelty requires PubChem, ChEMBL, SureChEMBL and a patent search, none of which run offline. Similarity to an active drug does not transfer that drug's activity, and distance from one is not evidence of anything.

The same simulation across the potency interval

Assigned IC50 (uM)	Growth inhibition %
0.0001	90.5
0.0027	88.5
0.0971	75.6
3.5388	9.1
129.0328	-4.5

What the uncertainty actually costs

The potency above is a prediction, not a measurement, and its 95% conformal interval spans a factor of 1,767,665 in IC50. Running the same tissue model across that interval gives growth inhibition anywhere from -4.5 % to 90.5 %, against the 75.6 % reported at the point estimate.

The potency interval admits tissue responses from -4.5 to 90.5 percent growth inhibition. That band includes no effect, so the simulation does not establish that this compound does anything.

Conformal prediction gives a set with 95% coverage, not a distribution, so no mean or spread over this band is quoted: the method does not say where inside the interval the truth lies.

Could a chemist make this?

SA score	Band	Alerts	Families	Verdict
4.29	demanding	0	none	no alerts, synthetically reachable

Substructure alerts and SA score are coarse heuristics with known false positives; approved drugs trip them. Passing means not obviously implausible, not that the compound can be made or works.

Potential, and what it rests on

Signal	Reading	What it is based on
Predicted potency	favourable	pIC50 7.01, calibrated qsar
Potency certainty	low by construction	conformal interval 3.1 log units
Simulated tissue effect	distinguishable	TGI 76.3 % over 5 seeds, p < 0.001
Toxicology alerts	favourable	risk index 8.4 of 100, ten category screen
Drug likeness	moderate	QED 0.533, Lipinski passes
Model applicability	inside domain	Tanimoto 0.66 to nearest training compound
Structural distinctness	Related to known chemistry	max Tanimoto 0.62 over 111 local structures
Medicinal chemistry	no alerts	SA 4.29 (demanding), PAINS, Brenk and NIH catalogues

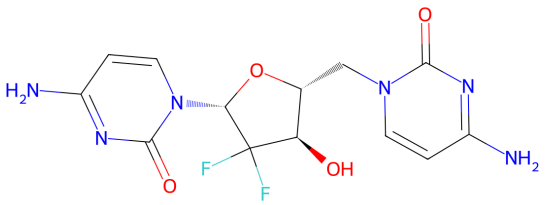
Not assessed at all

whether a synthetic route actually exists (SA score estimates difficulty, it does not plan a synthesis); chemical stability and reactivity under real conditions; absorption, distribution, metabolism and excretion in a living system; whether the compound reaches a pancreatic tumour through desmoplastic stroma; any experimental measurement whatsoever.

Verdict

This candidate separated from the negative control and survived correction for the number of candidates tested. That warrants further computational scrutiny and nothing more.

Candidate 05: Compound-A215BD



Identity

Canonical SMILES	Nc1ccn(C[C@H]2O[C@@H](n3ccc(N)nc3=O)C(F)(F)[C@@H]2O)c(=O)n1
InChIKey	RFQHSPXNBVMUFH-BDODKLCJSA-N
Molecular formula	C13H14F2N6O4
Murcko scaffold	O=c1ncccn1CC1CCC(n2cccnc2=O)O1
Origin	generation 20 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
356.3	-1.44	151.3	3	8	0.605	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	6.715	3.59 to 9.84	ordering signal
Implied IC50 (uM)	0.1929	spans orders of magnitude	not a quantity
Active probability	0.824	0.72 to 0.92	the stronger prediction
Applicability domain	0.78	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.671	0.000	0.605	0.916	0.783	0.651	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	67.79 % over 5 seeds
95% confidence interval	62.24 % to 73.34 %
Interval excludes zero	yes
Per seed values	68.6, 64.9, 61.9, 70.5, 73.2
Resistant clone fraction	0.849
Value used as selection objective	68.6 % on the search seed
Versus negative control	p < 0.001, Hedges g 8.49
Distinguishable, uncorrected	yes
Distinguishable, Bonferroni corrected	yes

Does something like this already exist?

Local verdict	Close analogue of a known compound
Nearest local structure	Gemcitabine (Tanimoto 0.783)
Exact match	none in the local corpus
Same skeleton	none
Shared Murcko scaffold	none
Corpus searched	111 local structures

Closest local structure	Source	Tanimoto	Same scaffold
Gemcitabine	EVOCURE compound library	0.783	no
Capecitabine	Curated reference dataset	0.260	no
Capecitabine	EVOCURE compound library	0.218	no
Linezolid	EVOCURE compound library	0.165	no
Linezolid	Curated reference dataset	0.165	no

This is not a prior art search

This searches roughly 120 local structures. It is a sanity check, not a prior art search. A structure absent here may still be published, catalogued or patented, and nothing in this result establishes that it is new. Confirming novelty requires PubChem, ChEMBL, SureChEMBL and a patent search, none of which run offline. Similarity to an active drug does not transfer that drug's activity, and distance from one is not evidence of anything.

The same simulation across the potency interval

Assigned IC50 (uM)	Growth inhibition %
0.0001	89.6
0.0053	90.4
0.1929	58.9
7.0330	10.5
256.4422	-4.5

What the uncertainty actually costs

The potency above is a prediction, not a measurement, and its 95% conformal interval spans a factor of 1,767,665 in IC50. Running the same tissue model across that interval gives growth inhibition anywhere from -4.5 % to 90.4 %, against the 58.9 % reported at the point estimate.

The potency interval admits tissue responses from -4.5 to 90.4 percent growth inhibition. That band includes no effect, so the simulation does not establish that this compound does anything.

Conformal prediction gives a set with 95% coverage, not a distribution, so no mean or spread over this band is quoted: the method does not say where inside the interval the truth lies.

Could a chemist make this?

SA score	Band	Alerts	Families	Verdict
4.10	demanding	0	none	no alerts, synthetically reachable

Substructure alerts and SA score are coarse heuristics with known false positives; approved drugs trip them. Passing means not obviously implausible, not that the compound can be made or works.

Potential, and what it rests on

Signal	Reading	What it is based on
Predicted potency	favourable	pIC50 6.71, calibrated qsar
Potency certainty	low by construction	conformal interval 3.1 log units
Simulated tissue effect	distinguishable	TGI 67.8 % over 5 seeds, p < 0.001
Toxicology alerts	favourable	risk index 8.4 of 100, ten category screen
Drug likeness	favourable	QED 0.605, Lipinski passes
Model applicability	inside domain	Tanimoto 0.78 to nearest training compound
Structural distinctness	Close analogue of a known	max Tanimoto 0.78 over 111 local structures
Medicinal chemistry	no alerts	SA 4.10 (demanding), PAINS, Brenk and NIH catalogues

Not assessed at all

whether a synthetic route actually exists (SA score estimates difficulty, it does not plan a synthesis); chemical stability and reactivity under real conditions; absorption, distribution, metabolism and excretion in a living system; whether the compound reaches a pancreatic tumour through desmoplastic stroma; any experimental measurement whatsoever.

Verdict

This candidate separated from the negative control and survived correction for the number of candidates tested. That warrants further computational scrutiny and nothing more.

3. What this run does not show

The value of a screening report lies as much in its boundaries as in its ranking. The following are not minor caveats; they determine what the numbers above can be used for.

No experimental evidence of any kind
No compound here has been synthesised. There is no cell assay, no animal model and no clinical data. Every number is the output of a model.

The tissue model is not calibrated to real tumours
Kinetics, diffusion constants, clone behaviour and drug response parameters are plausible but have not been fitted to experimental pancreatic tumour data. Simulated growth inhibition is a property of this software.

The potency model orders compounds, it does not quantify them
R2 of 0.70 under repeated grouped cross validation is real signal, but the conformal interval of about 3.1 log units still spans orders of magnitude in IC50. Use the ranking, not the numbers.

A small, narrow training set
43 curated compounds. Applicability domain is measured against that set, so high domain confidence means close to the training data, which also means structurally unadventurous.

Virtual time is not real time
Generations and exposure hours index the simulation. They do not correspond to chemical synthesis, biological timescales or clinical development.

Structural similarity is not efficacy and not novelty
Matching a known drug does not transfer its activity. Failing to match anything in a small local library does not make a structure new.

No ADME, formulation or synthesis assessment
Nothing here addresses whether these molecules can be made at reasonable cost, are stable, are orally available, or reach a pancreatic tumour in a living system. Dense desmoplastic stroma makes that last point a central problem in this disease.

Multiple comparisons
Five candidates were tested against one control arm. The per candidate threshold is uncorrected; a Bonferroni corrected threshold would be $p < 0.01$, and the results file records both.

Reference drugs and novel structures are not scored the same way
Gemcitabine's potency in the tissue model comes from curated experimental bioactivity. A newly generated structure has no measurement, so its potency comes from the structure-property model, which assigns micromolar values far weaker than the positive control's. At the dose used here that difference alone is close to sufficient to explain why no candidate matches the control. The comparison is therefore between a measurement and a prediction, not between two compounds, and failing to beat the control is weak evidence about the molecules.

The tissue objective is noisy and not independent
All arms in a generation share one seed, so selection sees a single draw of a quantity whose spread across seeds crosses zero. The objective also depends on the potency the pharmacology model assigns, so it is correlated with the activity objective rather than an independent check on it.

If you take one thing from this report
Pancreatic ductal adenocarcinoma has a five year survival rate in the low teens, and that is precisely why early computational output about it should be described carefully. This platform is a hypothesis generator. The appropriate next step for any compound listed here is scrutiny by a medicinal chemist, not optimism.

4. Reproducibility

4.1 Environment of this run

Platform	Linux-6.18.44-fc-v185-x86_64-with-glibc2.39
Python	3.13.17
Compute backend	NumPy CPU float64
Device	host CPU
Field precision	float64
Search seed	20261027
Replicate seeds	20261027, 20261028, 20261029, 20261030, 20261031
Results hash	19ed50d0d68e985a25b87765b9c766fc0f86d32775248f84

4.2 Library versions

Package	Version
numpy	2.5.4
rdkit	2026.9.1
scikit-learn	1.9.1
scipy	1.18.1
torch	not installed

4.3 Source file hashes, SHA-256

File	SHA-256 (first 40 characters)
data/reference/pdac_reference_dataset.json	39c4df1535728f1cda6f568d3f69e5b19f03c223
models/scientific/pdac_qsar_v1.joblib	d76d7996d3532fe88ad6d8c7bad3db0c01e88f83
simulation/biology/diseases/pdac.py	30b77458673ab40a9f19678400fae20d89e8e8fc
simulation/core/engine.py	0173527b2bee9a3eb6f654be62c73dbe0e30aabc
simulation/discovery/pipeline.py	02c5214ee8bc39d47a540565802c441091aeee40
simulation/discovery/workbench.py	0a24e6f8a979a6c494cfad3aedbc66f11fbb9cb7
simulation/pharmacology/admet.py	0542add5a260f3c19c71fbff74e60837fc7ca067

4.4 Reproducing this document

The platform is open source and so is this result. The reference dataset, the pipeline, the tissue models and the report generator are all in the repository. No model weights are required: the QSAR artifact is rebuilt from the committed dataset, and the optional language models were not used for any quantitative output.

```
python scripts/train_qsar.py

# Linux and macOS
EVOCURE_COMPUTE=cpu EVOCURE_DISABLE_LAYA_NEURAL=1 \
  python scripts/run_pdac_report_experiment.py

# Windows PowerShell
$env:EVOCURE_COMPUTE = "cpu"
$env:EVOCURE_DISABLE_LAYA_NEURAL = "1"
python scripts\run_pdac_report_experiment.py

python scripts/build_candidate_pdf.py
```

Forcing the CPU backend is what makes a run bit reproducible. GPU backends that support float64 agree to within 1e-12 on the substrate fields; float32 backends such as DirectML and Metal differ at around 1e-6, which is harmless for the conclusions and fatal for hash

equality. The backend used is recorded above and in the results file.

4.5 Licence and reuse

The platform is released under the MIT licence and the curated reference dataset under CC-BY 4.0. Results, including this report and the underlying JSON, are published so that the discoveries stay open: anyone may reproduce, check, contradict or build on them without permission.